IT5009 MULTIMEDIA TECHNOLOGIES

ASSIGNMENT - IV

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How does interactive video differ from traditional video content in terms of user experience and content flow? Explain how user input shapes the structure and outcome of interactive multimedia applications, with relevant examples.

Interactive video, also known as IV, is a form of digital video that supports user interaction. It provides the viewer the ability to click on a desktop or touch on mobile devices within the video for an action to occur. These clickable areas, or 'hotspots', can perform an action when clicked or touched.

For example, a video may

- → display additional information
- → jump to a different part of the video
- → change the storyline

On the other hand, we have traditional video, which is more linear, more passive, and more one-dimensional. The main difference between linear video/traditional video is that interactive video is just traditional video with some additional features. However, there is more nuance to it.

Traditional video provides passive user engagement, whereas interactive video is active. The viewer has more control in interactive video, and pretty much no control in traditional video. Due to the interactive nature, there is a high retention rate in interactive video.

BENEFITS OF INTERACTIVE VIDEO

Interactive video provides a better user experience and content flow when compared to traditional video. The different elements and attributes of interactive video are enhanced by the varying kinds of user input provided. Interactive video 'reacts' to what the user wants, making everything responsive and proving a seamless experience.

Enhanced User Engagement

In today's landscape, content needs to be not only informative, but also engaging to stand out. Studies show that 80% of online users will watch a video, but only 20% will read your content. Interactive videos therefore have a higher user engagement rate as they help users visualise benefits and go beyond the scope of passively consuming content.

Easy Customisations

Interactive videos can be tailor-made for a personalised experience. The user can make choices and design which helps keep them engaged and also helps with information retention. Unlike traditional videos, these videos help collect data and valuable insights, which helps businesses navigate customers better in the future.

Increased Conversions

A study has found that interactive videos help convert buyers 70% of the time. This is twice the amount as conversions from static content. For e-commerce and sales, interactive videos allow users to click on products, view details, and make purchases directly within the video, which also helps boost conversions. Such videos are rare to come by, and thus they ~~require~~ pique the user interest significantly as well.

Virality

Interactive videos not only help boost sales directly, but also indirectly by garnering more eyeballs on your business than ever. Since people on social media are always looking for fun and new types of content, when they come across a well-executed interactive video, they are more likely to share it. This gets businesses more views, followers, and comment engagement.

Choosing Between Traditional and Interactive Videos

Choose interactive video for

- → increasing engagement
- → boosting conversion rates
- → e-commerce and marketing
- → educational content
- → data collection

Choose traditional video for

- → brand awareness
- → narrative content
- → situations with budget constraints

CONCLUSION

Looking to the future, the need for interactive videos will only boom. More and more users want to participate in what they consume rather than passively watch or read. And while both traditional and interactive videos will continue to hold space in marketing strategies, one without the use of the other would be a disservice to a business' overall digital content strategy.

Storytelling, branding, and narrative-style content are a better fit for traditional videos, whereas personalise, action-driven experiences are better for interactive videos.

Design a Virtual Reality game aimed at teaching high school students about environmental conservation. Explain how VR elements and game mechanics can be integrated to create an engaging and interactive learning experience.

Anecdotal evidence from philanthropic fundraisers shows that VR increases empathy, and can influence people toward pro-environmental behaviour. In VR, users are immersed in situations intended to feel more like the real world through technology, such as 360° video viewed through 3D headsets that block out visual and auditory distractions. The framing of the message as either positive or negative has long shown to have an effect on behaviour, although consensus on the impact of framing has not been reached in relation to encouraging contributions to public goods.

Once of interest only to tech geeks and hardcore gamers, new VR apps can immerse anyone in lifelike virtual environments and experiences that can help raise ecological awareness and promote environmentally friendly behaviour in the real world.

The Stanford Ocean Acidification Experience was brought to more than 270 high school students, college students, and adults. In one such test, high school seniors in a marine biology class took on new virtual identities in the simulation. Each became a pink coral on a rocky underwater reef throbbing with urchins, bream, snails, and other creatures.

By the end of the simulation - which fast forwards to what the reef will look like at the end of this century - those brilliantly varied and colourful species have disappeared. They are replaced by slimy green algae and the silver Salema Porgy - a fish that will likely thrive in more acidic waters. Eventually, the viewer's virtual coral skeleton disintegrates.

The simulation was effective at making users feel a connection with their bodies, according to researchers who tracked the students' movements. Some of the students swiveled their heads and twisted their bodies during the simulation.

In all of the study's in-school experiments, participants demonstrated increasing knowledge and awareness about ocean acidification as their time in the VR learning environment grew longer.

One question that could potentially be asked is - why not just use a simple video and present it to an audience? The thing is, VR provides an immersion that no other existing technology can possibly provide.

To allow people to get immersed into a river and experience the impact of different pollution sources - this is something that, without VR, one can only talk about. VR not only provides visualisation - it provides a real-world feel at the comfort of one's home.

The significant aspects of VR that enable complete immersion for its users, providing interactivity at a level hitherto unheard of, are as follows:

360°- Degree View Perception :→ The ability to ~~feel~~ freely move within a 3D environment.
→ Motion sensors detect the user's movements and adjusts the view on the screen in real-time, creating the illusion of 'being there'.

Immersion: → Combination of
1) 3D graphics
2) Sound Effects
3) Other sensory inputs

work together to make players feel truly surrounded by the virtual environment.

~~Interactivity~~ → This multisensory experience adds depth and realism, further immersing players in the game world.

Interactivity: → Players can manipulate virtual objects, explore environments, and even interact with others users in real-time.

→ This level of interaction goes beyond what is possible with traditional media.

Visual Navigation: → Creating a sense of presence
→ Guiding players in the virtual world
→ Effective lighting, colour contrast, and strategic
→ ~~G~~ object placement helps guide players in the desired direction.

Audio Navigation: → Spatial audio plays a vital role in immersing players by placing sounds strategically in specific locations.

Mechanics: → Proper haptics
→ Smooth locomotion
→ Arm swing
→ Realistic motion

all provide a seamless and enjoyable experience which allow effortless navigation, while providing minimal discomfort and simulation sickness.

In conclusion, designing a captivating and immersive VR environment for an interactive learning experience requires careful integration of principles such as presence, interactivity, immersion, agency, feedback, realism, and challenge. By embracing these identified principles, the full potential of VR can be realised, enabling entirely new, unique, and objectively better experiences in various new and old areas.